LIBRARY MANAGEMENT SYSTEM

Python Midterm Project Report

Team members: Add the three names before submission.
Course: Programming Principles 2
Submission: October 18, 2026

Project overview

This is a small console program for managing a library. It stores books, members, and borrowing records in SQLite. A short menu lets the user add, display, search, update, and cancel records, then view simple counts or save a text report.

Database

Table	Fields	Purpose
books	book_id, title, author, genre, available	Stores catalog details and availability
members	member_id, name, email, program	Stores student members and study programs
borrowings	borrowing_id, book_id, member_id, returned	Connects books with members and tracks return status

The book_id and member_id fields in borrowings connect the three related tables. Sample member records are fictional.

How the program works

Classes

Book and **Member** store record details. The program uses these objects to build Pandas tables for the console.

Functions

Function	Action
add_book / add_member	Save a new book or member
borrow_book / delete_record	Create or cancel a borrowing record
display_records / search_records	Show books, members, borrowing records, or matching results
update_book	Change a book title or author
show_statistics	Count borrowing records and active loans; find the most borrowed book
export_results	Write counts, borrowing statistics, and the list of loans to results.txt

Python concepts from class

The code uses basic variables, input, output, conditions, and a while loop for the menu. Functions handle each action. Book and Member are simple classes. Pandas prints query results as tables. SQLite stores the data, and a text file saves results. Regular expressions check member email addresses; random assigns sample borrowing records.

Input checks

The program checks that required names are present, validates member email format, asks for number IDs, and prevents borrowing a book that is already out.

Sample data and results

Dataset

The included database has 50 varied book titles, 40 fictional student members, and 30 borrowing records. The catalog includes genre, and member records include a study program. Twelve loans are active and 18 are marked returned. The sample names and records are fictional.

Statistics

Calculation	Sample result
Total borrowing records	30
Books currently borrowed	12
Most borrowed book	Python Crash Course — 8 loans

Output from the running program

```
Choose an option: 9
Total borrowing records: 30
Books currently borrowed: 12
Most frequently borrowed book: Python Crash Course - 8 loans
```

Team contributions

Team member	Contribution
Member 1 - add name	Add the work completed and presentation part.
Member 2 - add name	Add the work completed and presentation part.
Member 3 - add name	Add the work completed and presentation part.

Run with python3 main.py. Choose option 10 to create or refresh results.txt.